CUSTOMER EXPERIENCE

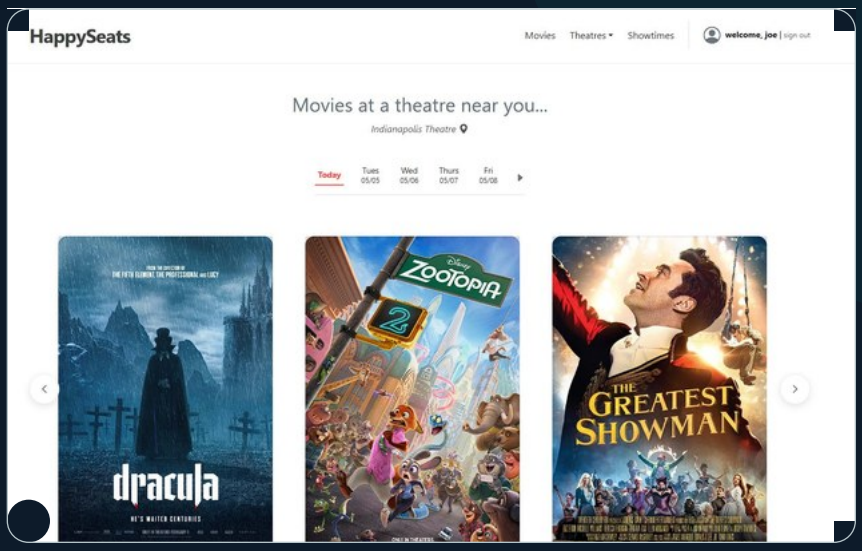
HappySeats

Powered by SeatSphere

Your seat. Your movie. Your moment.

- **Browse 60+ live showtimes**
- **Interactive seat selection**
- **ADA-accessible seat mapping**
- **Instant QR ticket generation**
- **Secure login & booking history**
- **Real-time availability updates**

Spring Boot | Node.js | MySQL



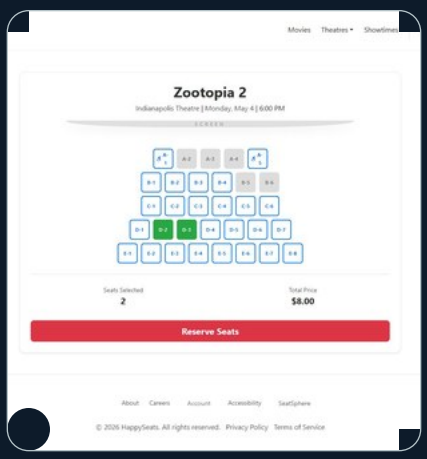
Live movie catalog with real-time showtimes

Demo

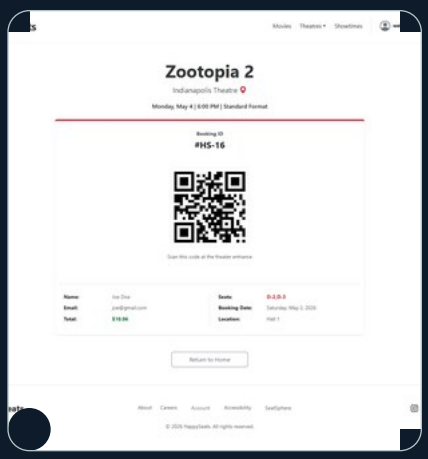
https://iu.mediaspace.kaltura.com/media/t/1_jov5jta3

Project Site

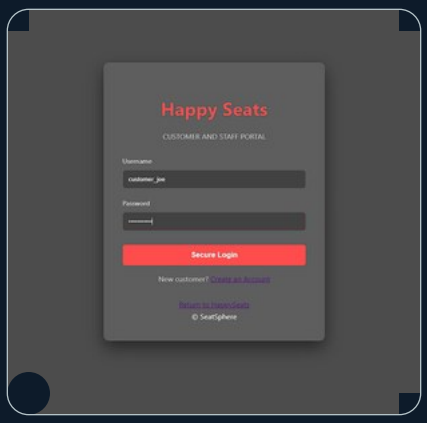
https://github.com/daniellexkemp/SeatSphere



Interactive seat map



Auto-generated QR ticket



Secure login portal

Easy account creation

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Developed by Danielle Kemp.
Indiana University. INFO-C 450 System Design.

SeatSphere

Theater Management & Admin Dashboard

MANAGER PORTAL • ROLE-SECURED

SEATSPHERE

HappySeats Movies Tickets welcome, danielle | sign out

Bookings

Live entries from the SeatSphere database.

Booking ID	Movie	Showtime	Total Price	Ticket Codes	
#1	Zootopia 2	06:00 PM	$14.34	TKT-R8S32R TKT-R1NGAY TKT-TQPLBN	Delete
#2	Dracula	05:00 PM	$4.00	TKT-DRAC11	Delete
#3	Cosmic Drift	05:30 PM	$12.50	TKT-COSMIC	Delete
#4	Zootopia 2	06:00 PM	$10.06	TKT-PZ191H TKT-Z764LR	Delete
#5	Star Wars: A New Hope	06:15 PM	$49.65	TKT-T04VL9 TKT-UN3UBV TKT-9U9I36	Delete
#6	The Greatest Showman	08:00 PM	$22.90	TKT-9OMHX2 TKT-N9Y851 TKT-BESX8U TKT-52GB78 TKT-P4AWTK	Delete
#7	Inception	03:30 PM	$27.18	TKT-712C8N TKT-AEVL3B	Delete
#8	The Batman	10:30 AM	$29.32	TKT-U5PZTO TKT-P8BVAQ	Delete
#9	Cosmic Drift	11:00 AM	$14.88	TKT-5IPN2G	Delete
#10	K-Pop Demon Hunter	07:30 PM	$18.62	TKT-VUM1Z8 TKT-RQJ3JN	Delete
#11	Toy Story 5	02:30 PM	$65.70	TKT-SDQ0D7 TKT-1MDHHH TKT-UVYZV4 TKT-XATAT3 TKT-X885R7 TKT-0ZSR90	Delete

Live bookings dashboard — accessible only to MANAGER / ADMIN-role staff via Spring Security

Relational Engine

Seat availability calculated in real-time via dynamic SQL joins. No stale flags — every page fires a live lookup against showtime_id in the MySQL schema.

1

Role-Based Security

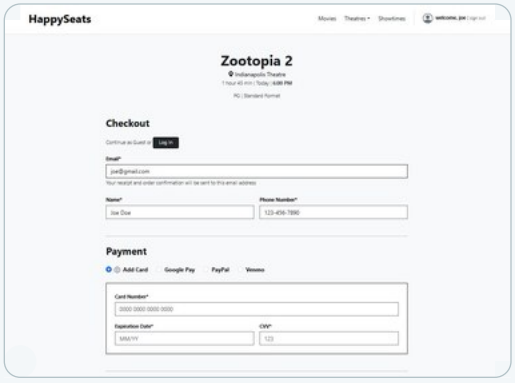
Spring Security enforces strict separation. Customers access HappySeats; only staff with the MANAGER role can view or delete booking records in the dashboard.

2

Transactional Integrity

Each purchase triggers a multi-table transaction: booking record, user_id link, total_amount calculation, and unique QR code string — all generated atomically.

3



Checkout & payment flow

SeatSphere × HappySeats

Built with Spring Boot · Node.js · MySQL · JavaScript

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